

TYUKHTET, Russian Federation

WMO: 294560

Lat: 56.536N

Lon: 89.317E

Elev: 205

StdP: 98.89

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.4	-35.0	-42.1	0.1	-39.3	-37.6	0.1	-34.7	8.5	-8.5	7.5	-10.7	0.7	230	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	28.8	18.9	27.1	18.0	25.4	17.4	20.6	25.8	19.5	24.9	18.5	23.7	2.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.0	23.5	17.5	12.9	22.1	16.5	12.0	21.1	60.7	25.7	56.7	25.1	53.3	23.9	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.7	5.1	-40.7	31.9	5.0	1.7	-44.3	33.1	-47.2	34.1	-50.0	35.1	-53.6	36.3		
(5)			-40.5	21.7	4.1	2.1	-43.5	23.3	-46.0	24.5	-48.3	25.7	-51.3	27.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-18.1	-12.9	-5.6	3.0	9.6	16.4	18.5	16.1	8.6	1.0	-10.1	-13.3
	DBStd	13.94	9.40	7.58	6.59	4.90	5.60	4.54	3.32	3.70	4.69	5.42	8.75	8.78
	HDD10.0	3993	871	642	484	216	81	7	0	3	82	282	604	722
	HDD18.3	6361	1129	876	743	459	276	86	39	88	293	537	854	980
	CDD10.0	769	0	0	0	6	67	200	262	193	39	2	0	0
	CDD18.3	96	0	0	0	0	4	28	43	19	1	0	0	0
	CDH23.3	894	0	0	0	2	53	324	351	153	12	0	0	0
	CDH26.7	193	0	0	0	0	9	79	78	27	1	0	0	0
Wind	WSAvg	2.2	2.3	2.3	2.5	2.7	2.6	1.9	1.7	1.8	2.0	2.3	2.4	2.5
Precipitation	PrecAvg	530	24	20	24	30	42	66	62	73	50	53	47	39
	PrecMax	661	41	39	60	52	107	118	111	165	90	91	72	68
	PrecMin	412	3	7	5	7	14	15	20	30	4	21	26	12
	PrecStd	61	9	8	13	11	22	24	24	30	22	15	13	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	1.5	8.6	20.6	28.2	31.2	30.9	29.1	25.1	16.9	7.9	1.8
		MCWB	-0.7	-1.1	3.5	10.7	16.4	18.8	21.0	19.7	15.9	9.6	5.5	0.0
	2%	DB	-1.7	-0.2	6.4	16.7	24.7	29.0	28.7	26.9	21.3	12.9	4.6	0.2
		MCWB	-3.0	-2.4	2.2	8.9	14.5	18.0	19.7	19.5	14.5	8.0	2.6	-1.1
	5%	DB	-3.8	-1.4	4.8	13.8	22.2	26.9	26.9	24.9	18.8	10.4	2.6	-1.6
		MCWB	-4.8	-2.9	1.1	7.4	13.2	17.1	18.8	18.3	13.1	7.4	1.1	-2.8
	10%	DB	-5.5	-2.9	3.2	11.3	19.5	24.9	25.2	23.0	16.2	8.2	0.7	-3.3
		MCWB	-6.5	-4.2	0.2	5.9	12.1	16.3	18.0	17.0	11.7	5.7	-0.4	-4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	-0.4	4.2	11.6	17.5	21.0	22.6	21.8	16.2	10.6	5.7	0.5
		MCDB	1.2	0.9	7.2	19.5	26.4	27.1	28.8	27.0	22.5	13.7	7.7	1.2
	2%	WB	-2.8	-1.9	2.8	9.6	15.4	19.2	21.0	20.1	14.9	9.1	3.3	-1.1
		MCDB	-1.8	-0.7	5.7	14.8	22.3	26.6	26.2	25.0	21.1	12.5	4.2	0.0
	5%	WB	-4.8	-2.9	1.6	8.0	14.1	18.2	19.9	18.8	13.6	7.4	1.1	-2.6
		MCDB	-3.9	-1.3	4.0	13.5	20.9	25.0	25.1	23.6	17.9	10.1	2.5	-1.5
	10%	WB	-6.6	-4.3	0.3	6.1	12.6	17.3	19.0	17.6	12.3	5.8	-0.3	-4.4
		MCDB	-5.6	-2.9	2.9	10.7	18.7	22.8	23.9	21.9	15.6	8.2	0.6	-3.4
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.9	9.1	9.0	10.1	10.6	9.7	9.5	8.5	6.1	6.0	6.4
		MCDBR	6.9	6.6	10.0	12.7	14.0	14.4	12.7	12.3	13.6	9.7	5.8	6.0
		MCWBR	6.3	5.5	6.9	7.0	7.0	5.7	5.4	5.8	7.3	6.1	4.8	5.4
	5% WB	MCDBR	6.9	6.6	8.0	11.9	12.2	12.1	10.8	10.7	12.2	8.8	5.3	5.8
		MCWBR	6.4	5.9	5.9	7.3	6.5	5.4	5.4	5.7	6.9	5.8	4.6	5.5
Clear-Sky Solar Irradiance	taub	0.290	0.322	0.370	0.398	0.408	0.407	0.433	0.409	0.372	0.351	0.319	0.290	
	taud	1.986	1.966	1.907	2.064	2.202	2.294	2.220	2.270	2.321	2.218	2.024	1.935	
	Ebn at Noon	547	691	754	808	828	836	800	795	771	668	500	422	
	Edh at Noon	66	102	141	141	131	121	128	114	94	80	64	54	
All-Sky Solar Radiation	RadAvg	0.74	1.70	3.02	4.24	4.93	5.88	5.35	4.23	2.73	1.38	0.67	0.43	
	RadStd	0.10	0.15	0.23	0.34	0.55	0.55	0.37	0.33	0.36	0.13	0.10	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page